# IBM Cognos Analytics-Project

## *Case Study 9*

***Industry:*** **Gym & Fitness**

***Modules:*** **Reporting, Exploration, Dashboard, StoryMaking**

## FitZone Gym - Member Retention & Revenue Analytics


| | | |
|---|---|---|
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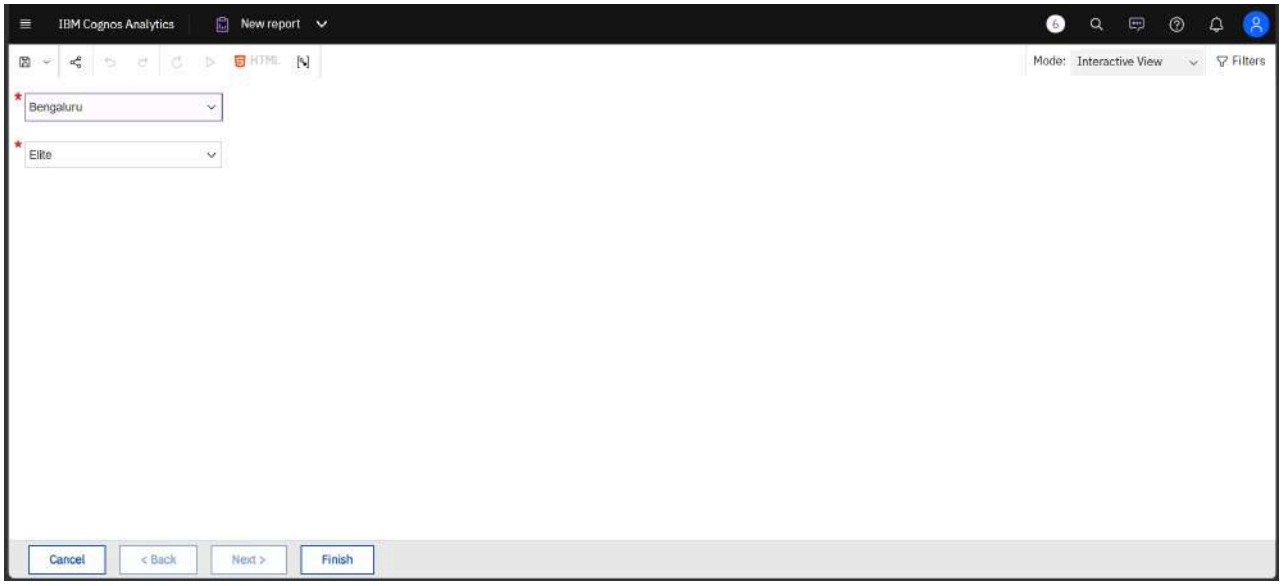
## Q1 [Reporting]

**Create a Membership Report grouped by Branch and Membership Type showing Active Members, Expired Members, Renewal Rate %, and Dropout Count. Apply Conditional Styling: branches with Dropout Count > 50 in Red, 25–50 in Amber, < 25 in Green. Add Header with FitZone branding, Footer with page number and report date. Add prompts for Branch City and Membership Type.**

- Interactive Report Prompts for Branch City and Membership Type to enable customized data filtering. Branch-wise Membership Report featuring FitZone branding and detailed conditional styling for dropout analysis



## Q2 [Reporting]

**Build an At-Risk Member Report listing members whose class attendance is below 4 sessions/month AND whose membership expires within 30 days. Add columns for Days to Expiry (calculated) and Risk Level (High/Medium based on attendance). Apply Conditional Styling for risk levels. Summarize total at-risk members per branch in the footer.**

***At-Risk Member Analytics Report - FitZone***

| Member Name | Branch Name | Classes Attended | Expiry Date | Days to Expiry | Risk Level |
|---|---|---|---|---|---|
| Member_143 | FitZone Andheri | 0 | May 10, 2024 | -695 | High Risk |
| Member_141 | | | Sep 5, 2024 | -577 | High Risk |
| Member_101 | | 1 | Oct 29, 2024 | -523 | High Risk |
| Member_164 | | 2 | Apr 7, 2025 | -363 | Medium Risk |
| Member_195 | | 3 | Sep 1, 2024 | -581 | Medium Risk |
| Member_081 | FitZone Anna Nagar | 0 | Jul 20, 2024 | -624 | High Risk |
| Member_018 | | | Nov 10, 2024 | -511 | High Risk |
| Member_181 | | | Feb 28, 2025 | -401 | High Risk |
| Member_043 | | | Aug 20, 2025 | -228 | High Risk |
| Member_146 | | 1 | Apr 30, 2024 | -705 | High Risk |
| Member_092 | | 2 | Sep 23, 2024 | -559 | Medium Risk |
| Member_131 | | | Oct 22, 2024 | -530 | Medium Risk |
| Member_044 | | | Sep 22, 2025 | -195 | Medium Risk |
| Member_178 | | 3 | Apr 26, 2024 | -709 | Medium Risk |
| Member_105 | | | Oct 5, 2024 | -547 | Medium Risk |
| Member_107 | | | Oct 31, 2024 | -521 | Medium Risk |
| Member_037 | | | Dec 29, 2024 | -462 | Medium Risk |
| Member_047 | FitZone Banjara Hills | 0 | Aug 7, 2024 | -606 | High Risk |
| Member_132 | | 1 | Jun 4, 2024 | -670 | High Risk |
| Member_167 | | | Oct 16, 2024 | -536 | High Risk |
| Member_063 | | 2 | Aug 24, 2024 | -589 | Medium Risk |
| Member_188 | | | Nov 20, 2024 | -501 | Medium Risk |

1

*Visualizing members with low attendance and upcoming expiry dates using conditional formatting*

*Calculating days to expiry and summarizing total at-risk members per branch for management review*

## At-Risk Member Analytics Report - FitZone

| Member Name | Branch Name | Classes Attended | Expiry Date | Days to Expiry | Risk Level |
|---|---|---|---|---|---|
| Member_162 | FitZone Powai | 0 | Sep 20, 2024 | -562 | High Risk |
| Member_170 | | | Jan 31, 2025 | -429 | High Risk |
| Member_179 | | 1 | Sep 14, 2024 | -568 | High Risk |
| Member_142 | | 2 | Sep 1, 2024 | -581 | Medium Risk |
| Member_128 | | | Sep 29, 2024 | -553 | Medium Risk |
| Member_008 | | | Oct 13, 2024 | -539 | Medium Risk |
| Member_004 | FitZone Salt Lake | 0 | Sep 14, 2024 | -568 | High Risk |
| Member_073 | | | Sep 26, 2024 | -556 | High Risk |
| Member_034 | | | Oct 4, 2024 | -548 | High Risk |
| Member_104 | | | Nov 3, 2024 | -518 | High Risk |
| Member_096 | | | Nov 13, 2024 | -508 | High Risk |
| Member_088 | | | May 5, 2025 | -335 | High Risk |
| Member_157 | | 1 | Jun 4, 2024 | -670 | High Risk |
| Member_023 | | | Jul 14, 2024 | -630 | High Risk |
| Member_134 | | | Dec 31, 2024 | -460 | High Risk |
| Member_111 | | 2 | Jul 12, 2024 | -632 | Medium Risk |
| Member_079 | | | Apr 6, 2025 | -364 | Medium Risk |
| Member_013 | | | Apr 11, 2025 | -359 | Medium Risk |
| Member_171 | | 3 | May 17, 2024 | -688 | Medium Risk |
| Member_177 | | | Nov 27, 2024 | -494 | Medium Risk |
| Member_118 | | | Aug 14, 2025 | -234 | Medium Risk |

# *At-Risk Member Analytics Report - FitZone*

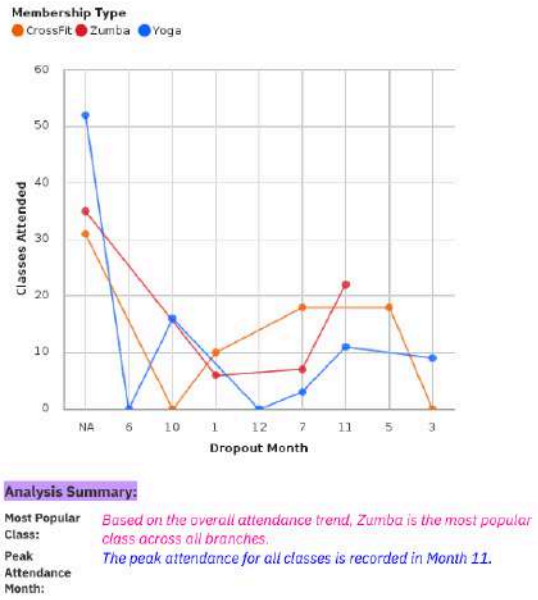
| Member Name | Branch Name | Classes Attended | Expiry Date | Days to Expiry | Risk Level |
|---|---|---|---|---|---|
| **64** | **Overall - Count** | | | | |

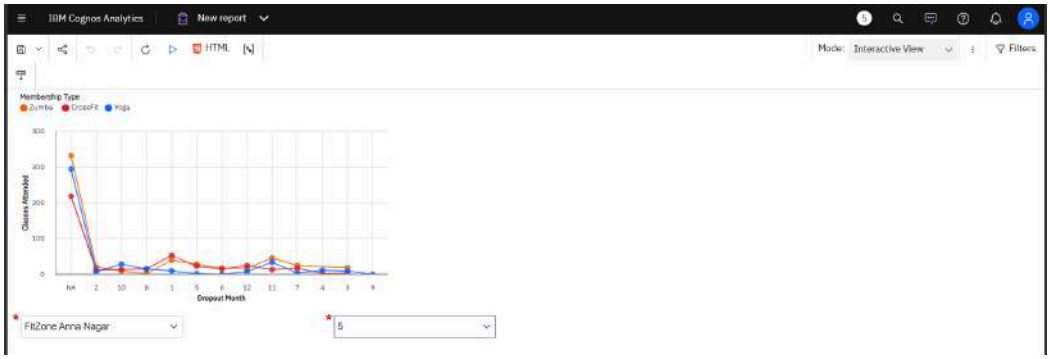
Apr 5, 2026

## Q3 [Reporting]

**Using Report Visualize Prompt, allow Branch Managers to select a chart type to view Monthly Class Attendance Trend for each class type (Zumba, Yoga, CrossFit). Add a Branch and Month filter prompt. Summarize the most popular class and peak attendance month. Use Blue for Yoga, Orange for Zumba, Red for CrossFit in Conditional Styling columns.**

- Interactive visualization allowing managers to toggle chart types for Monthly Class Attendance Trends.



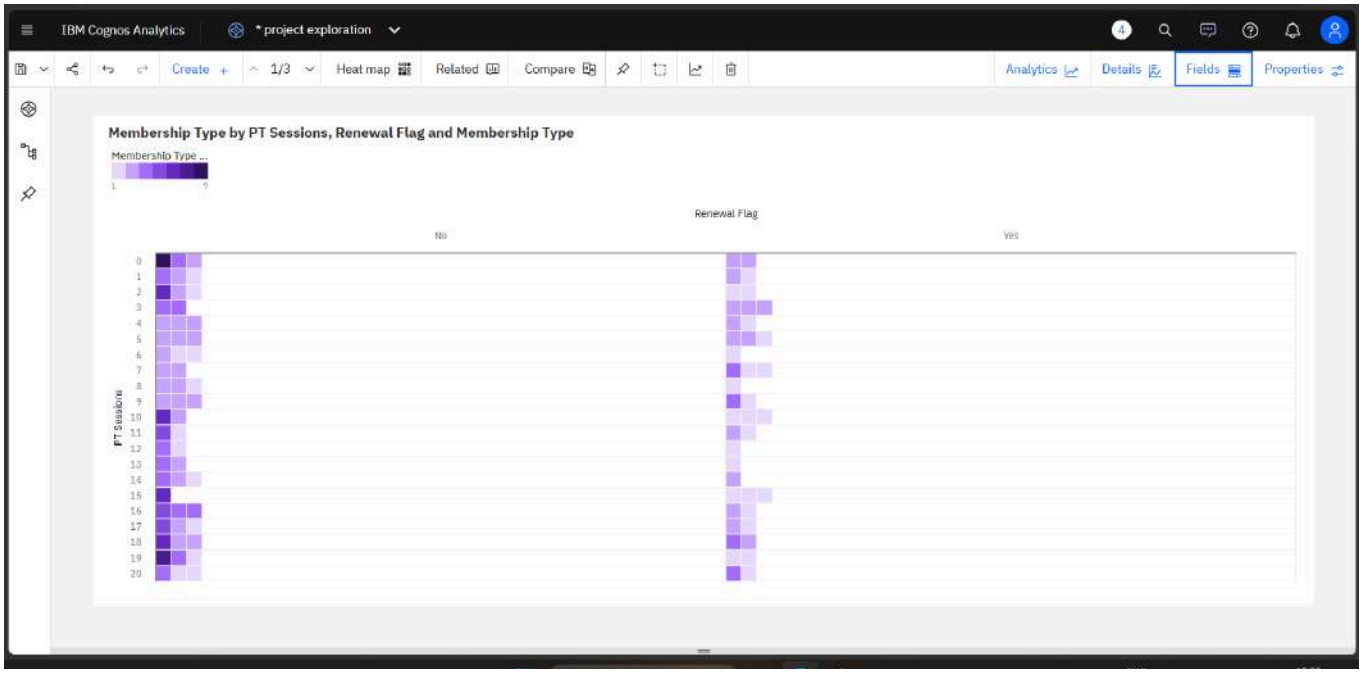

- Branch and Month filters with a summary highlighting peak attendance and color-coded class categories.

## Q4 [Exploration]

Use Cognos Exploration to analyze the FitZone member dataset. Explore whether Members with Personal Training sessions have higher renewal rates than those without.Investigate the relationship between Age Group, Gender, and Membership Type preference. Identify common characteristics of members who drop out within the first 3 months.

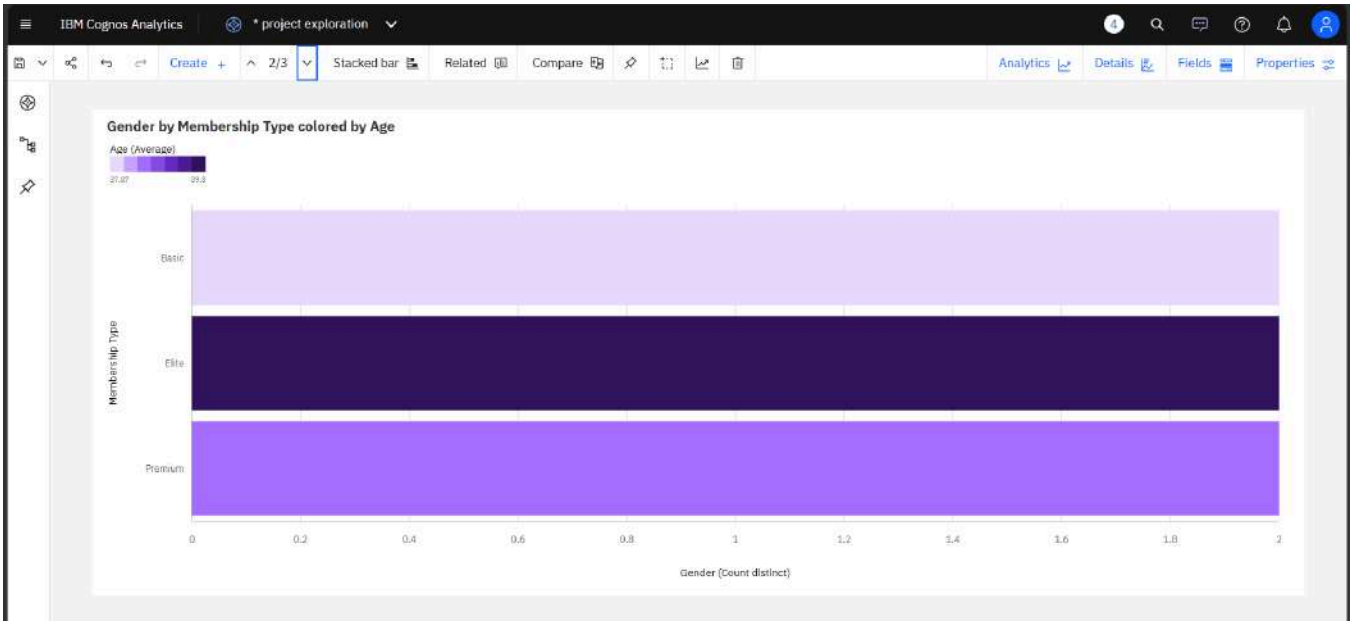
## PT Sessions vs Renewal Rate:

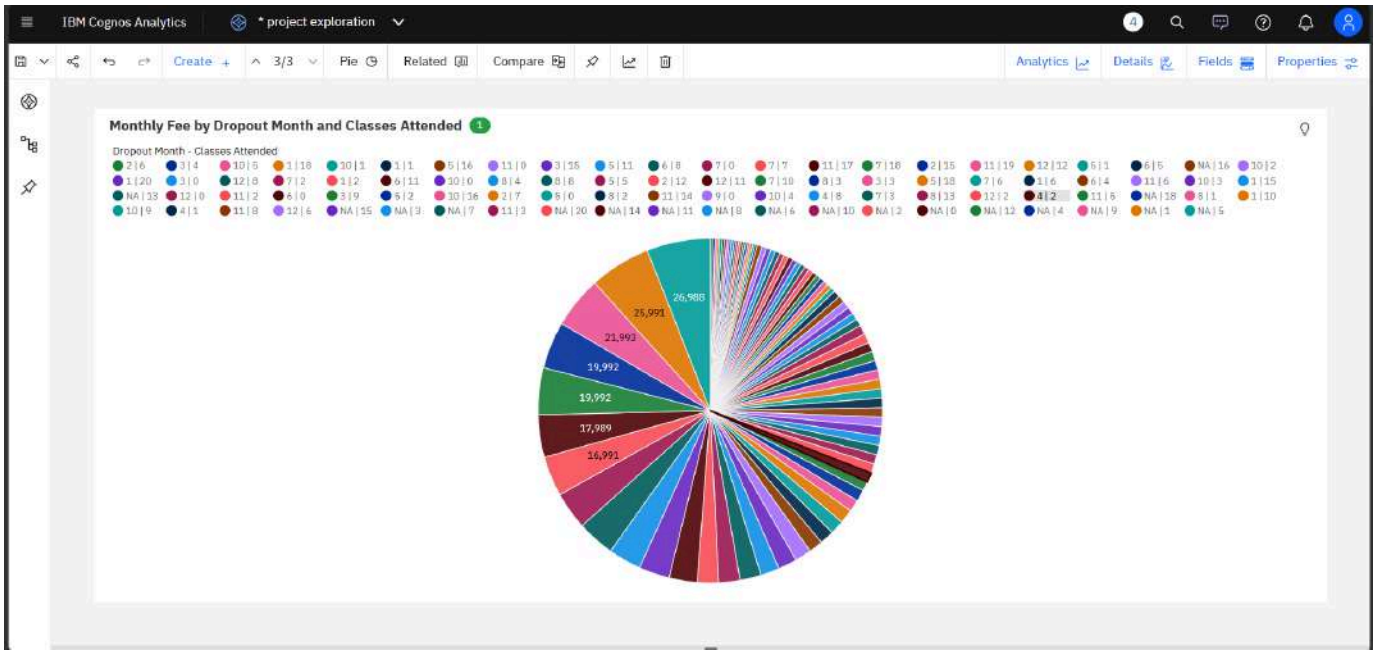
Analysis of Personal Training sessions and their impact on Membership Renewal.




## Member Preferences:

Relationship between Age Group, Gender, and Membership Type preference.



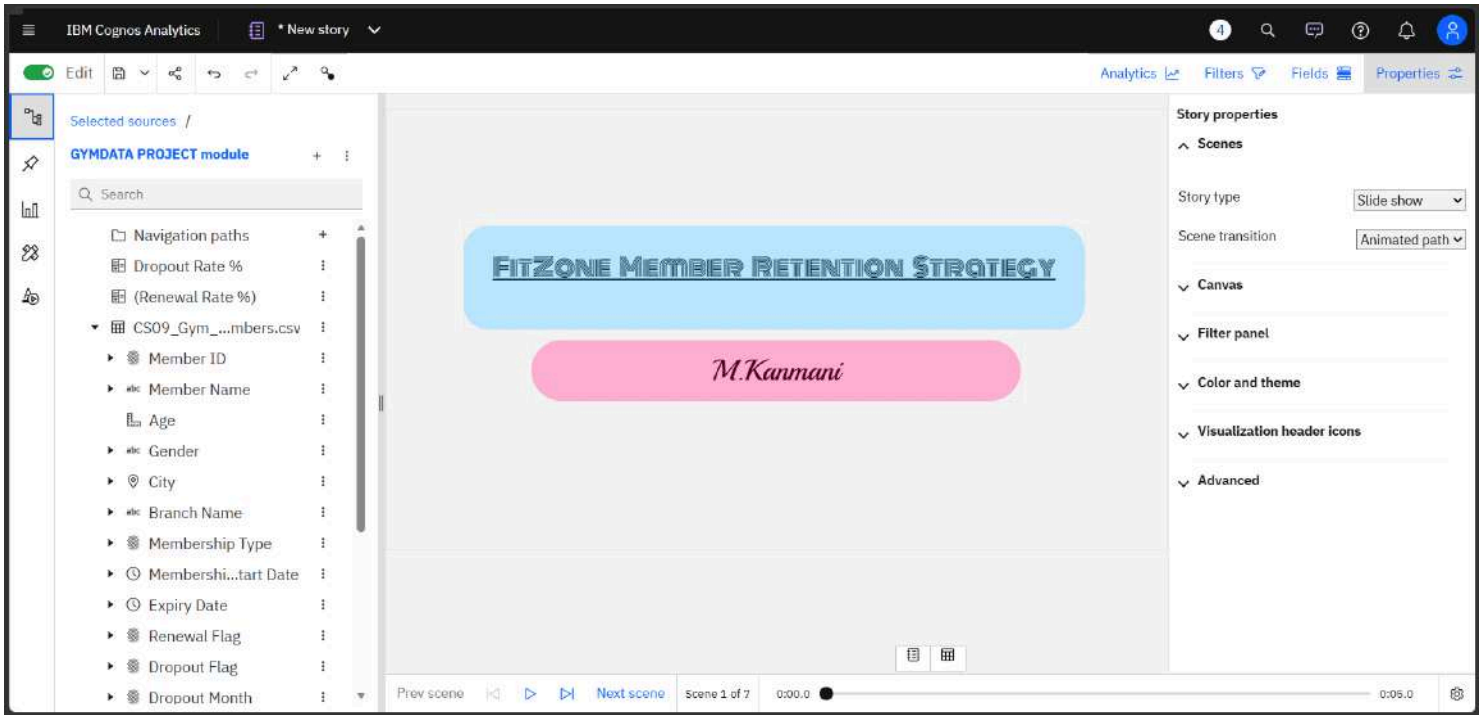

## **Early Dropout Analysis:**

Common characteristics of members who drop out within the first 3 months.

## Q5[Story Making]

**Create a Cognos Story for FitZone's Member Retention Strategy Meeting: Scene1 — Business Overview & Active Member Count, Scene 2 — Branch-wise Membership Performance, Scene 3 — Class Attendance Patterns & Popular Programs, Scene 4 — At-Risk Member Profile Analysis, Scene 5 — Trainer Performance & Impact on Renewals, Scene 6 —Retention Campaign Recommendations.**

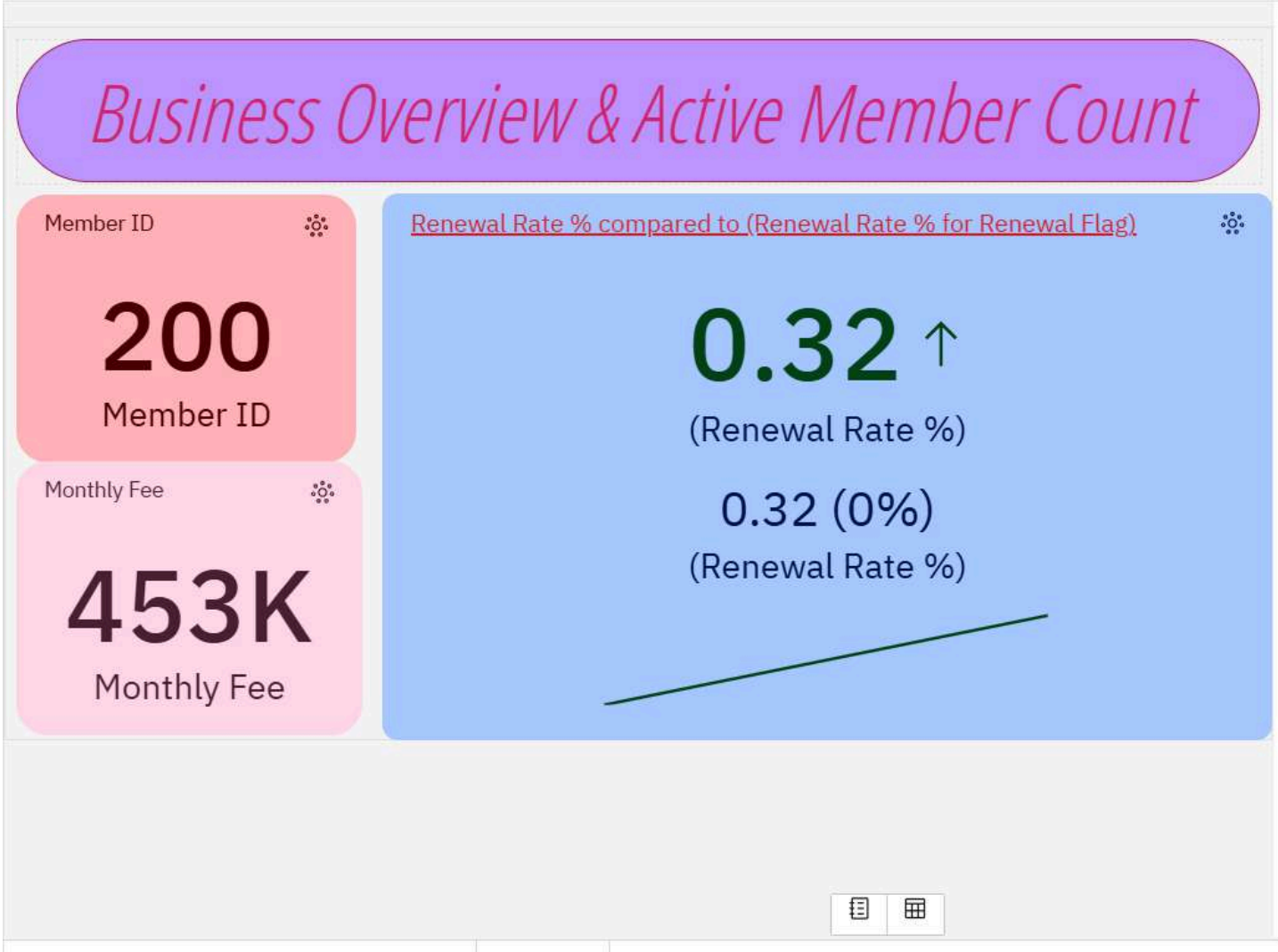
## Scene 1




## Scene 2: Business Overview

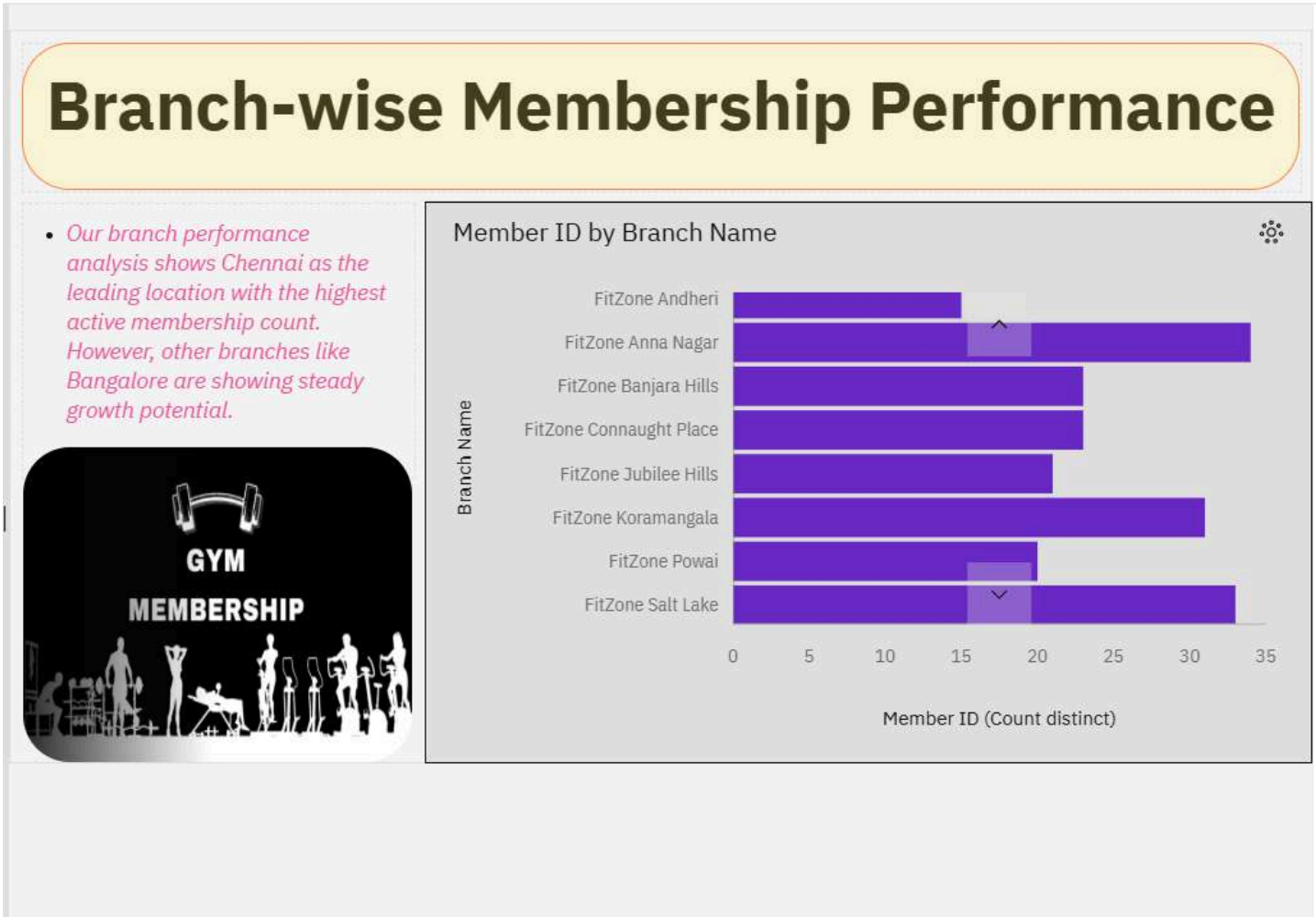
- Key Performance Indicators (KPIs) showcasing 200 active members and $453K total revenue, supported

by a renewal trend visualization.



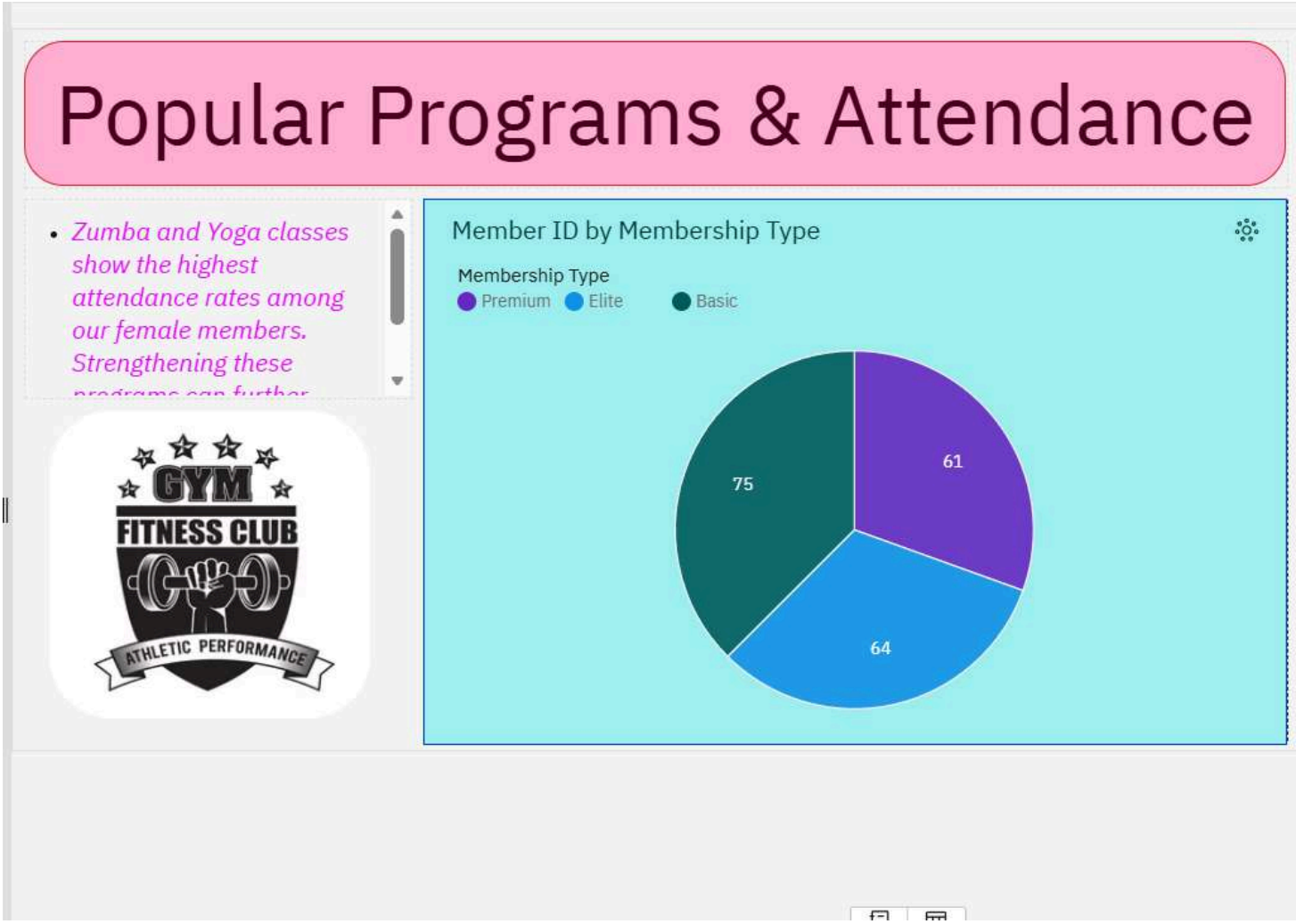

## Scene 3: Branch Performance

- Bar chart analysis evaluating membership performance across different branches to identify high-performing locations.




## Scene 4: Popular Programs

- Pie chart distribution illustrating class attendance patterns and identifying the most popular fitness programs.




## Scene 5: At-Risk Analysis

- Critical analysis of the 68% dropout rate, identifying at-risk members for early intervention and retention strategies.

## Scene 6: Trainer Performance

- Evaluation of trainer performance and their direct impact on member engagement and long-term retention.

# Trainer Performance & Retention

*High-performing trainers create better engagement, directly increasing renewal chances for their clients.*

Trainer Name

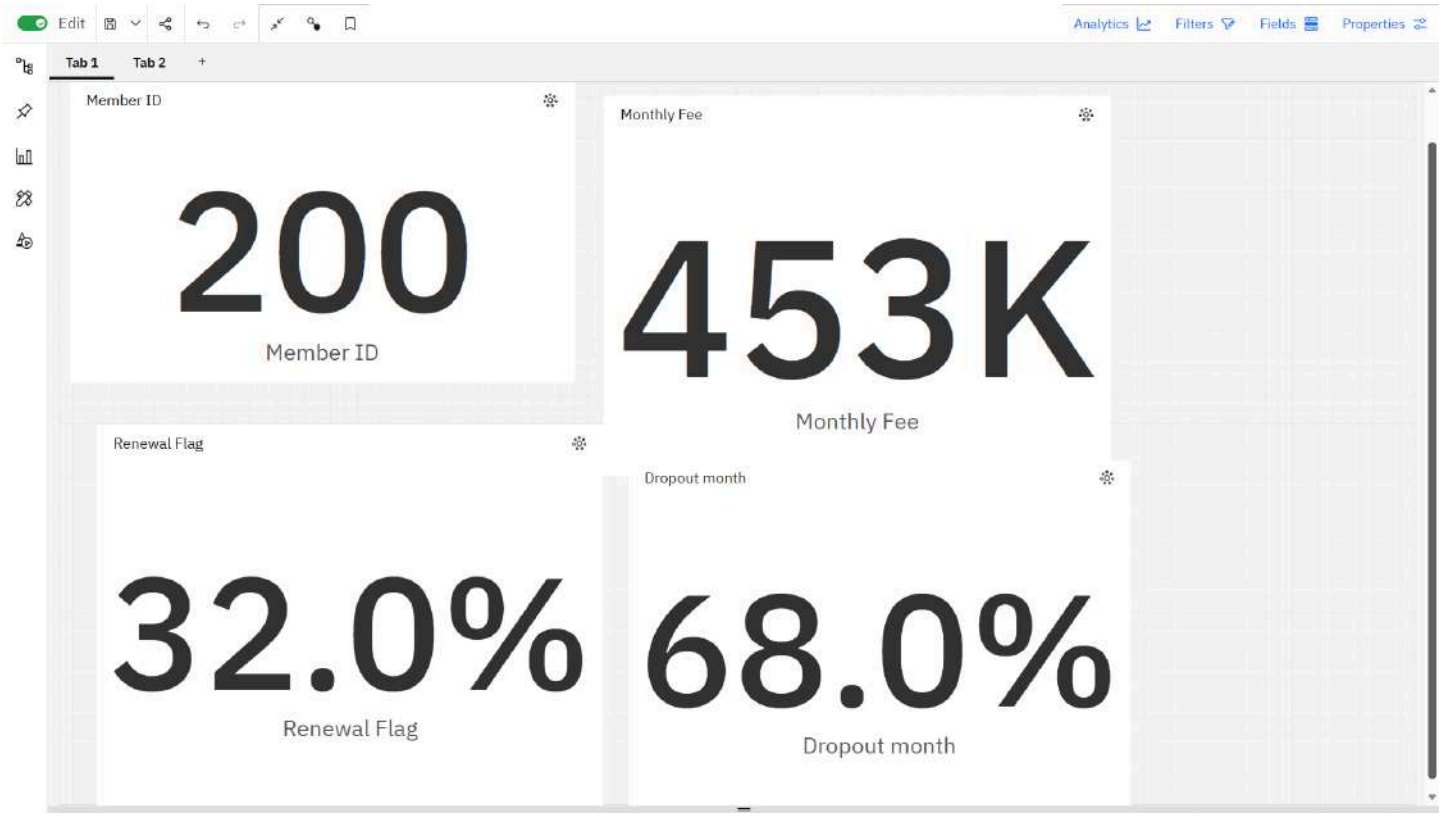
| Trainer Name |
| --- |
| Trainer_Arjun |
| Trainer_Kiran |
| Trainer_Priya |
| Trainer_Ravi |

## Q6 [Dashboard]

**Design a FitZone Operations Dashboard with: (1) KPI tiles — Total Active Members, Renewal Rate %, Monthly Revenue, Dropout Rate %, (2) A Bar chart for Members by Branch, (3) A Line chart for Monthly New Enrollments, (4) A Pie chart for Membership Type Distribution, (5) A table of Top 5 Trainers by Client Count. Add Branch and Month filter widgets.**

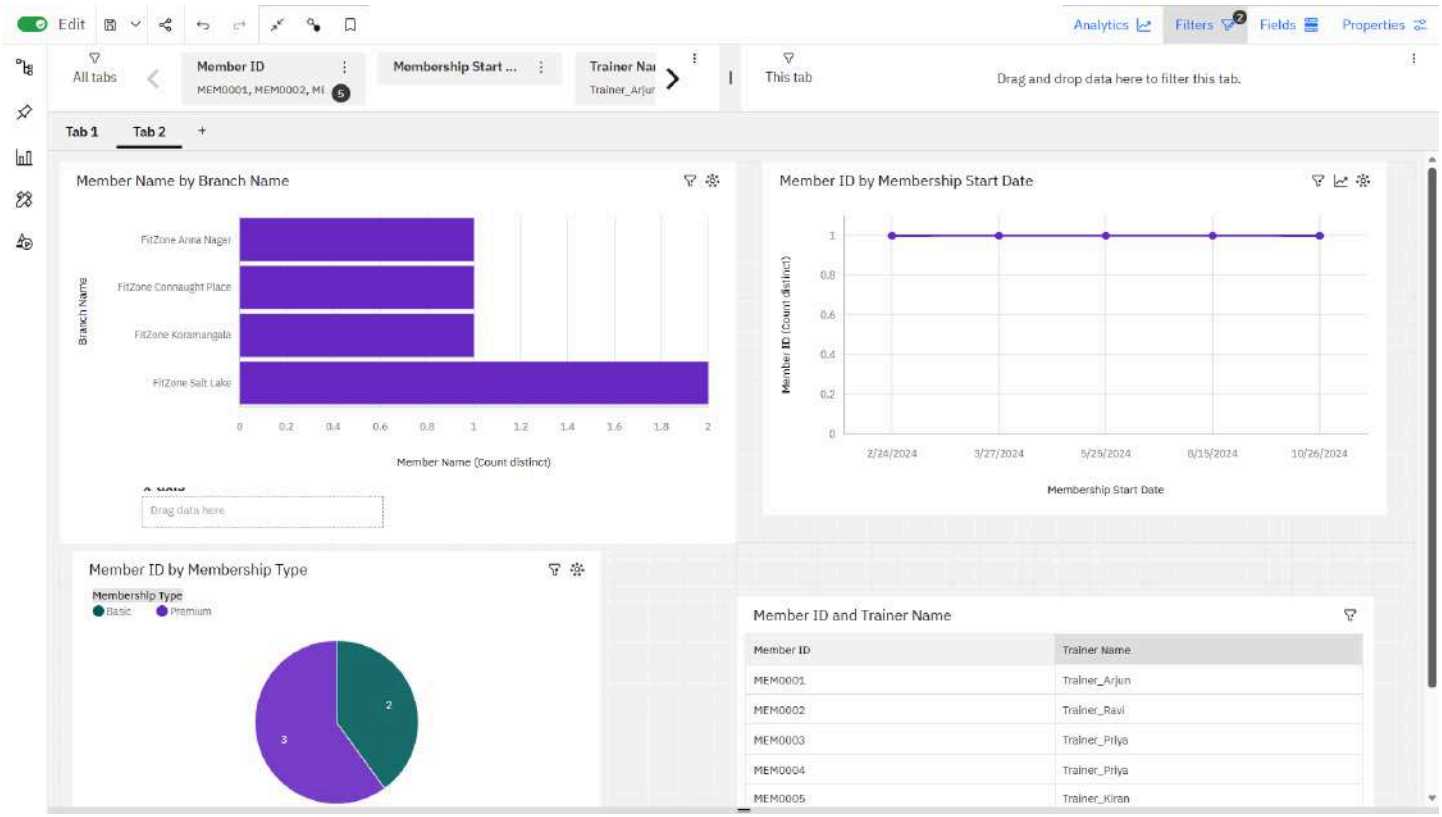
### Tab 1 :(KPI Summary)

- **Total Active Members:** 200 (Member ID count).
- **Monthly Revenue:** 453K (Sum of Monthly Fee).
- **Renewal Rate %:** 32.0%
- **Dropout Rate %:** 68.0% (Dropout month calculation).



## Tab 2: (Operational Visuals)

- **Bar Chart:** "Member Name by Branch Name" (Purple bars)
- **Line Chart:** "Member ID by Membership Start Date" (Trend line)
- **Pie Chart:** "Member ID by Membership Type" (Premium vs Basic)
- **Table:** "Member ID and Trainer Name" (Top Trainers list)
- **Filter Widgets:** Top-la "Member ID" and "Membership Start Date" widgets.



# FitZone Web Portal: Built with HTML, CSS & JavaScript

## CODING:

```
<!DOCTYPE html>

<html lang="en">

<head>

   <meta charset="UTF-8">

   <meta name="viewport" content="width=device-width, initial-scale=1.0, user-scalable=yes">

   <title>FitZone Elite Analytics Portal | Complete Solution</title>

   <link href="https://fonts.googleapis.com/css2?family=Inter:opsz,wght@14..32,300;14..32,400;14..32,500;14..32,600;14..32,700;14..32,800&display=swap" rel="stylesheet">

   <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-beta3/css/all.min.css">

   <script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>

   <script src="https://cdn.sheetjs.com/xlsx-0.20.2/package/dist/xlsx.full.min.js"></script>

   <script src="https://cdnjs.cloudflare.com/ajax/libs/html2pdf.js/0.10.1/html2pdf.bundle.min.js"></script>

   <style>

      * {

         margin: 0;

         padding: 0;

         box-sizing: border-box;

      }
```

```
body {
    font-family: 'Inter', sans-serif;
    background: linear-gradient(135deg, #0a2a28 0%, #0d3d3a 100%);
    min-height: 100vh;
}

/* ========== LOGIN SECTION - BEAUTIFUL VERSION ========== */
.login-wrapper {
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
    width: 100%;
    padding: 20px;
    background: linear-gradient(135deg, #0a2a28 0%, #0d3d3a 100%);
}

.login-card {
    background: white;
    border-radius: 32px;
    width: 100%;
    max-width: 480px;
    padding: 48px 40px;
    box-shadow: 0 30px 60px rgba(0, 0, 0, 0.2);
    transition: transform 0.3s ease;
    animation: fadeInUp 0.6s ease;
```

```
}

@keyframes fadeInUp {
    from {
        opacity: 0;
        transform: translateY(30px);
    }
    to {
        opacity: 1;
        transform: translateY(0);
    }
}

.login-card:hover {
    transform: translateY(-5px);
}

.logo-section {
    text-align: center;
    margin-bottom: 36px;
}

.logo-icon {
    width: 80px;
    height: 80px;
    background: linear-gradient(135deg, #1e7e6c, #0f5c4e);
```

```
    border-radius: 24px;
    display: flex;
    align-items: center;
    justify-content: center;
    margin: 0 auto 20px;
    box-shadow: 0 10px 25px rgba(30,126,108,0.3);
    animation: pulse 2s infinite;
}

@keyframes pulse {
    0%, 100% { transform: scale(1); }
    50% { transform: scale(1.05); }
}

.logo-icon i {
    font-size: 40px;
    color: white;
}

.logo-section h1 {
    font-size: 28px;
    font-weight: 800;
    color: #1e2a2e;
    margin-bottom: 8px;
}
```

```
.logo-section p {
    color: #6c8a87;
    font-size: 14px;
    font-weight: 500;
}

.input-field {
    margin-bottom: 24px;
}

.input-field label {
    display: block;
    font-size: 13px;
    font-weight: 600;
    color: #2c4a47;
    margin-bottom: 8px;
}

.input-field input {
    width: 100%;
    padding: 14px 16px;
    border: 1.5px solid #e0ecea;
    border-radius: 16px;
    font-size: 15px;
    transition: all 0.2s;
    background: #fafdfc;
```

```
}

.input-field input:focus {
    outline: none;
    border-color: #1e7e6c;
    box-shadow: 0 0 0 3px rgba(30,126,108,0.1);
    background: white;
}

.btn-signin {
    width: 100%;
    padding: 14px;
    background: linear-gradient(135deg, #1e7e6c, #146b5a);
    color: white;
    border: none;
    border-radius: 16px;
    font-size: 16px;
    font-weight: 700;
    cursor: pointer;
    transition: all 0.2s;
    margin-top: 12px;
}

.btn-signin:hover {
    transform: translateY(-2px);
    box-shadow: 0 8px 20px rgba(30,126,108,0.3);
```

```
}

.demo-info {
    text-align: center;
    margin-top: 28px;
    padding: 14px;
    background: #f0f7f5;
    border-radius: 16px;
    font-size: 13px;
    color: #5c7e7a;
}

.demo-info i {
    margin-right: 6px;
    color: #1e7e6c;
}

/* ========== DASHBOARD SECTION - SUPER VERSION ========== */
.dashboard-container {
    display: none;
    width: 100%;
    min-height: 100vh;
    background: #f5faf9;
}

body.dark-mode .dashboard-container {
```

```
    background: #121212;
}

/* Navbar */
.navbar {
    background: white;
    padding: 16px 32px;
    display: flex;
    justify-content: space-between;
    align-items: center;
    box-shadow: 0 2px 12px rgba(0, 0, 0, 0.04);
    position: sticky;
    top: 0;
    z-index: 100;
    flex-wrap: wrap;
    gap: 15px;
}

body.dark-mode .navbar {
    background: #1e1e1e;
    border-bottom: 1px solid #333;
}

.nav-brand {
    display: flex;
    align-items: center;
```

```
    gap: 12px;
}

.nav-brand i {
    font-size: 28px;
    color: #1e7e6c;
}

.nav-brand span {
    font-size: 22px;
    font-weight: 800;
    color: #1e2a2e;
}

body.dark-mode .nav-brand span {
    color: #e0e0e0;
}

.nav-actions {
    display: flex;
    align-items: center;
    gap: 16px;
    flex-wrap: wrap;
}

.action-btn {
```

```
    background: #f0f7f5;
    border: none;
    padding: 8px 16px;
    border-radius: 30px;
    cursor: pointer;
    font-weight: 500;
    font-size: 13px;
    transition: all 0.2s;
    display: flex;
    align-items: center;
    gap: 8px;
}

.action-btn:hover {
    transform: translateY(-2px);
    background: #e0ecea;
}

.user-badge {
    display: flex;
    align-items: center;
    gap: 10px;
    background: #f0f7f5;
    padding: 8px 18px;
    border-radius: 40px;
}
```

```
.user-badge i {
    font-size: 18px;
    color: #1e7e6c;
}

.logout-nav {
    background: #ffebee;
    border: none;
    color: #c62828;
    font-weight: 600;
    cursor: pointer;
    display: flex;
    align-items: center;
    gap: 6px;
    padding: 8px 16px;
    border-radius: 40px;
    transition: background 0.2s;
}

.logout-nav:hover {
    background: #ffcdd2;
}

/* Main Content */
.main-content {
```

```
    padding: 32px;
    max-width: 1400px;
    margin: 0 auto;
}

/* Stats Grid */
.stats-grid {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 24px;
    margin-bottom: 32px;
}

.stat-card {
    background: white;
    padding: 24px 20px;
    border-radius: 24px;
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.04);
    transition: all 0.2s;
    border: 1px solid #e6f0ee;
}

body.dark-mode .stat-card {
    background: #1e1e1e;
    border-color: #333;
}
```

```
.stat-card:hover {
    transform: translateY(-3px);
    box-shadow: 0 8px 24px rgba(0, 0, 0, 0.08);
}

.stat-title {
    font-size: 13px;
    font-weight: 600;
    text-transform: uppercase;
    letter-spacing: 1px;
    color: #6c8a87;
    margin-bottom: 12px;
}

.stat-value {
    font-size: 36px;
    font-weight: 800;
    color: #1e3a38;
}

body.dark-mode .stat-value {
    color: #48a999;
}

/* Filter Panel */
```

```
.filter-panel {
    background: white;
    padding: 20px;
    border-radius: 24px;
    margin-bottom: 32px;
    display: flex;
    gap: 15px;
    flex-wrap: wrap;
    align-items: flex-end;
    border: 1px solid #e6f0ee;
}

body.dark-mode .filter-panel {
    background: #1e1e1e;
    border-color: #333;
}

.filter-group {
    display: flex;
    flex-direction: column;
    gap: 6px;
}

.filter-group label {
    font-size: 12px;
    font-weight: 600;
```

```
    color: #6c8a87;
}

.filter-group input, .filter-group select {
    padding: 10px 14px;
    border: 1px solid #dcebe8;
    border-radius: 12px;
    font-size: 13px;
    background: white;
}

body.dark-mode .filter-group input,
body.dark-mode .filter-group select {
    background: #2d2d2d;
    border-color: #444;
    color: #e0e0e0;
}

/* Charts Grid */
.charts-grid {
    display: grid;
    grid-template-columns: repeat(2, 1fr);
    gap: 24px;
    margin-bottom: 32px;
}
```

```
.chart-card {
    background: white;
    padding: 20px;
    border-radius: 24px;
    border: 1px solid #e6f0ee;
}

body.dark-mode .chart-card {
    background: #1e1e1e;
    border-color: #333;
}

.chart-title {
    font-size: 16px;
    font-weight: 700;
    color: #1e3a38;
    margin-bottom: 16px;
    display: flex;
    align-items: center;
    gap: 8px;
}

body.dark-mode .chart-title {
    color: #e0e0e0;
}
```

```
.chart-title i {
    color: #1e7e6c;
}

canvas {
    max-height: 260px;
    width: 100%;
}

/* Alert Bar */
.alert-bar {
    background: linear-gradient(135deg, #fff9e6, #fff3d6);
    border-left: 4px solid #ff9800;
    padding: 16px 24px;
    border-radius: 16px;
    margin-bottom: 32px;
    display: flex;
    align-items: center;
    gap: 12px;
}

body.dark-mode .alert-bar {
    background: #2a2416;
    border-left-color: #ff9800;
}
```

```
.alert-bar i {
    font-size: 22px;
    color: #ff9800;
}

.alert-bar span {
    font-size: 14px;
    color: #5d4a1a;
}

body.dark-mode .alert-bar span {
    color: #ffcc80;
}

/* Member Table */
.table-container {
    background: white;
    border-radius: 24px;
    padding: 20px;
    margin-bottom: 32px;
    border: 1px solid #e6f0ee;
    overflow-x: auto;
}

body.dark-mode .table-container {
    background: #1e1e1e;
```

```
    border-color: #333;
}

.table-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 20px;
    flex-wrap: wrap;
    gap: 12px;
}

.table-header h3 {
    font-size: 18px;
    font-weight: 700;
    color: #1e3a38;
}

body.dark-mode .table-header h3 {
    color: #e0e0e0;
}

table {
    width: 100%;
    border-collapse: collapse;
}
```

```
th, td {
    padding: 14px 12px;
    text-align: left;
    border-bottom: 1px solid #edf3f1;
}

body.dark-mode th,
body.dark-mode td {
    border-bottom-color: #333;
    color: #d0d0d0;
}

th {
    background: #f8fcfb;
    font-weight: 600;
    font-size: 13px;
    color: #4a6e6a;
}

body.dark-mode th {
    background: #2a2a2a;
    color: #aaa;
}

tr:hover {
```

```
    background: #fafdfc;
    cursor: pointer;
}

body.dark-mode tr:hover {
    background: #2a2a2a;
}

.risk-badge {
    background: #ffebee;
    color: #c62828;
    padding: 4px 10px;
    border-radius: 30px;
    font-size: 12px;
    font-weight: 600;
    display: inline-block;
}

.active-badge {
    background: #e0f2f1;
    color: #00695c;
    padding: 4px 10px;
    border-radius: 30px;
    font-size: 12px;
    font-weight: 600;
}
```

```
/* Iframe */
.iframe-wrapper {
    background: white;
    border-radius: 24px;
    overflow: hidden;
    border: 1px solid #e6f0ee;
}

body.dark-mode .iframe-wrapper {
    background: #1e1e1e;
    border-color: #333;
}

iframe {
    width: 100%;
    height: 480px;
    border: none;
}

footer {
    text-align: center;
    padding: 24px;
    font-size: 12px;
    color: #8aaba7;
}
```

```
        @media (max-width: 1024px) {
            .stats-grid { grid-template-columns: repeat(2, 1fr); }
            .charts-grid { grid-template-columns: 1fr; }
            .main-content { padding: 20px; }
        }

        @media (max-width: 640px) {
            .stats-grid { grid-template-columns: 1fr; }
            .navbar { flex-direction: column; }
            .nav-actions { justify-content: center; }
        }
    </style>
</head>
<body>

<!-- BEAUTIFUL LOGIN PAGE -->
<div class="login-wrapper" id="loginWrapper">
    <div class="login-card">
        <div class="logo-section">
            <div class="logo-icon">
                <i class="fas fa-dumbbell"></i>
            </div>
            <h1>FitZone Portal</h1>
            <p>Elite Analytics Suite</p>
        </div>
```

```
      <form id="loginForm">
        <div class="input-field">
          <label><i class="fas fa-envelope" style="margin-right: 6px;"></i> Email</label>
          <input type="email" id="loginEmail" value="admin@fitzone.com"
autocomplete="off">
        </div>
        <div class="input-field">
          <label><i class="fas fa-lock" style="margin-right: 6px;"></i> Password</label>
          <input type="password" id="loginPassword" value="fitzone2025">
        </div>
        <button type="submit" class="btn-signin">
          <i class="fas fa-arrow-right-to-bracket" style="margin-right: 8px;"></i> Sign In
        </button>
      </form>
      <div class="demo-info">
        <i class="fas fa-chart-line"></i> demo: admin@fitzone.com / fitzone2025
      </div>
    </div>
</div>


<!-- SUPER DASHBOARD -->
<div class="dashboard-container" id="dashboardContainer">
    <div class="navbar">
      <div class="nav-brand">
        <i class="fas fa-heartbeat"></i>
        <span>FitZone · INSIGHT HUB</span>
```

```
    </div>
    <div class="nav-actions">
        <button class="action-btn" id="darkModeToggle"><i class="fas fa-moon"></i> Dark
Mode</button>
        <button class="action-btn" id="exportExcel"><i class="fas fa-file-excel"></i>
Excel</button>
        <button class="action-btn" id="exportPdf"><i class="fas fa-file-pdf"></i> PDF</button>
        <div class="user-badge">
            <i class="fas fa-user-circle"></i>
            <span id="userName">Admin</span>
        </div>
        <button class="logout-nav" id="logoutBtn">
            <i class="fas fa-sign-out-alt"></i> Exit
        </button>
    </div>
  </div>


  <div class="main-content">
    <!-- Stats Cards -->
    <div class="stats-grid" id="statsGrid"></div>


    <!-- Filter Panel -->
    <div class="filter-panel">
        <div class="filter-group">
            <label><i class="fas fa-search"></i> Search Member</label>
            <input type="text" id="searchMember" placeholder="ID or Name...">
        </div>
```

```
<div class="filter-group">
  <label><i class="fas fa-city"></i> City</label>
  <select id="cityFilter">
    <option value="">All Cities</option>
    <option>Mumbai</option><option>Delhi</option><option>Bengaluru</option>
    <option>Hyderabad</option><option>Chennai</option><option>Kolkata</option>
  </select>
</div>
<div class="filter-group">
  <label><i class="fas fa-tag"></i> Membership</label>
  <select id="membershipFilter">
    <option value="">All Types</option>
    <option>Basic</option><option>Premium</option><option>Elite</option>
  </select>
</div>
<div class="filter-group">
  <label><i class="fas fa-exclamation-triangle"></i> Risk Status</label>
  <select id="riskFilter">
    <option value="">All</option>
    <option value="high">High Risk</option>
    <option value="active">Active Only</option>
  </select>
</div>
<button id="resetFilters" class="action-btn" style="background:#e0ecea;"><i class="fas fa-undo-alt"></i> Reset</button>
</div>
```

```
<!-- Alert Bar -->
<div class="alert-bar" id="alertBar">
  <i class="fas fa-bell"></i>
  <span id="alertMessage">Loading insights...</span>
</div>

<!-- Charts Row -->
<div class="charts-grid">
  <div class="chart-card">
    <div class="chart-title"><i class="fas fa-chart-pie"></i> Membership Distribution</div>
    <canvas id="membershipChart"></canvas>
  </div>
  <div class="chart-card">
    <div class="chart-title"><i class="fas fa-city"></i> Members by City</div>
    <canvas id="cityChart"></canvas>
  </div>
</div>

<div class="charts-grid">
  <div class="chart-card">
    <div class="chart-title"><i class="fas fa-star"></i> Trainer Satisfaction</div>
    <canvas id="trainerChart"></canvas>
  </div>
  <div class="chart-card">
```

```
        <div class="chart-title"><i class="fas fa-chart-line"></i> Monthly Revenue Trend</div>
        <canvas id="revenueChart"></canvas>
      </div>
    </div>

    <!-- Member Table -->
    <div class="table-container">
      <div class="table-header">
        <h3><i class="fas fa-users"></i> Member Directory <span style="font-size:12px; font-weight:normal;">(click row for profile)</span></h3>
      </div>
      <div style="overflow-x: auto;">
        <table id="memberTable">
          <thead>
            <tr><th>Member ID</th><th>Name</th><th>City</th><th>Membership</th><th>Satisfaction</th><th>Status</th><th>Trainer</th></tr>
          </thead>
          <tbody id="memberTableBody"></tbody>
        </table>
      </div>
    </div>

    <!-- IBM Cognos Iframe -->
    <div class="iframe-wrapper">
      <iframe src="https://ap2.ca.analytics.ibm.com/bi/?perspective=story&amp;pathRef=.my_folders%2Fstory%2Bmaking%2Bproject&amp;closeWindowOnLastView=true&amp;ui_appbar=false&amp;ui_navbar=false&amp;shareMode=embedd
```

```
ed&amp;action=view&amp;sceneId=model0000019d66a99f56_00000002&amp;sceneTime=5000&amp;nav_filter=true" title="Cognos Dashboard"></iframe>

      </div>


      <footer>© 2025 FitZone Fitness | Elite Analytics Platform | Real-time Member Intelligence</footer>

   </div>

</div>


<script>

   // ========== GENERATE FULL DATASET (200 Members) ==========

   const fullMembers = [];

   const citiesList = ['Mumbai', 'Delhi', 'Bengaluru', 'Hyderabad', 'Chennai', 'Kolkata'];

   const membershipTypes = ['Basic', 'Premium', 'Elite'];

   const trainersList = ['Trainer_Arjun', 'Trainer_Ravi', 'Trainer_Priya', 'Trainer_Kiran', 'Trainer_Sunita'];


   for (let i = 1; i <= 200; i++) {

      let city = citiesList[Math.floor(Math.random() * citiesList.length)];

      let membership = membershipTypes[Math.floor(Math.random() * 3)];

      let trainer = trainersList[Math.floor(Math.random() * 5)];

      let satisfaction = +(Math.random() * 3.5 + 1).toFixed(1);

      let dropoutFlag = Math.random() > 0.7 ? "Yes" : "No";

      let monthlyFee = membership === 'Basic' ? 999 : membership === 'Premium' ? 1999 : 3999;

      let additionalRevenue = Math.random() * 10000;
```

```
    fullMembers.push({
        id: `MEM${String(i).padStart(4, '0')}`,
        name: `Member_${String(i).padStart(3, '0')}`,
        city: city,
        membership: membership,
        satisfaction: satisfaction,
        dropoutFlag: dropoutFlag,
        trainer: trainer,
        monthlyFee: monthlyFee,
        additionalRevenue: additionalRevenue,
        classesEnrolled: Math.floor(Math.random() * 5) + 1,
        classesAttended: Math.floor(Math.random() * 20)
    });
}

// Realistic distribution
for (let i = 0; i < 55; i++) fullMembers[i].city = "Mumbai";
for (let i = 55; i < 105; i++) fullMembers[i].city = "Hyderabad";

let currentFiltered = [...fullMembers];

// Chart variables
let membershipChart, cityChart, trainerChart, revenueChart;

// ========== UPDATE DASHBOARD ==========
function updateDashboard() {
```

```
let total = currentFiltered.length;

let active = currentFiltered.filter(m => m.dropoutFlag === "No").length;

let avgSat = (currentFiltered.reduce((a, b) => a + b.satisfaction, 0) / total).toFixed(1);

let totalRev = currentFiltered.reduce((s, m) => s + m.monthlyFee + (m.additionalRevenue || 0), 0);


// Update Stats

document.getElementById('statsGrid').innerHTML = `

  <div class="stat-card"><div class="stat-title"><i class="fas fa-users"></i> Total Members</div><div class="stat-value">${total}</div></div>

  <div class="stat-card"><div class="stat-title"><i class="fas fa-user-check"></i> Active Members</div><div class="stat-value">${active}</div></div>

  <div class="stat-card"><div class="stat-title"><i class="fas fa-star"></i> Avg Satisfaction</div><div class="stat-value">${avgSat} / 5</div></div>

  <div class="stat-card"><div class="stat-title"><i class="fas fa-rupee-sign"></i> Total Revenue</div><div class="stat-value">₹${Math.round(totalRev).toLocaleString()}</div></div>

`;


// Update Alert

let lowSat = currentFiltered.filter(m => m.satisfaction < 2).length;

let highRisk = currentFiltered.filter(m => m.dropoutFlag === "Yes" && m.satisfaction < 2.5).length;

document.getElementById('alertMessage').innerHTML = `  ${lowSat} members have low satisfaction (&lt;2) |  ${highRisk} members at high dropout risk |  ${currentFiltered.filter(m => m.satisfaction >= 4.5).length} happy promoters!`;


// Membership Distribution

let memDist = { Basic: 0, Premium: 0, Elite: 0 };

currentFiltered.forEach(m => memDist[m.membership]++);
```

```
// City Distribution
let cityDist = {};
currentFiltered.forEach(m => cityDist[m.city] = (cityDist[m.city] || 0) + 1);

// Trainer Satisfaction
let trainerSat = {};
currentFiltered.forEach(m => {
    if (!trainerSat[m.trainer]) trainerSat[m.trainer] = { sum: 0, count: 0 };
    trainerSat[m.trainer].sum += m.satisfaction;
    trainerSat[m.trainer].count++;
});
let trainerAvg = {};
for (let [k, v] of Object.entries(trainerSat)) {
    trainerAvg[k] = (v.sum / v.count).toFixed(1);
}

// Update Charts
if (membershipChart) membershipChart.destroy();
membershipChart = new Chart(document.getElementById('membershipChart'), {
    type: 'doughnut',
    data: { labels: ['Basic', 'Premium', 'Elite'], datasets: [{ data: [memDist.Basic,
memDist.Premium, memDist.Elite], backgroundColor: ['#1e7e6c', '#48a999', '#80cbc4'] }] }
});

if (cityChart) cityChart.destroy();
```

```
    cityChart = new Chart(document.getElementById('cityChart'), {
        type: 'bar',
        data: { labels: Object.keys(cityDist), datasets: [{ label: 'Members', data: Object.values(cityDist), backgroundColor: '#1e7e6c', borderRadius: 8 }] }
    });

    if (trainerChart) trainerChart.destroy();
    trainerChart = new Chart(document.getElementById('trainerChart'), {
        type: 'bar',
        data: { labels: Object.keys(trainerAvg), datasets: [{ label: 'Avg Satisfaction', data: Object.values(trainerAvg), backgroundColor: '#ff9800' }] }
    });

    if (revenueChart) revenueChart.destroy();
    revenueChart = new Chart(document.getElementById('revenueChart'), {
        type: 'line',
        data: { labels: ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'], datasets: [{ label: 'Revenue (₹ Lakhs)', data: [12.4, 13.2, 14.5, 13.8, 15.2, 16.8], borderColor: '#1e7e6c', tension: 0.3, fill: true, backgroundColor: 'rgba(30,126,108,0.1)' }] }
    });

    // Update Table
    let tbody = document.getElementById('memberTableBody');
    tbody.innerHTML = currentFiltered.slice(0, 50).map(m => `
        <tr onclick="showMemberProfile('${m.id}')">
            <td>${m.id}</td><td>${m.name}</td><td>${m.city}</td><td>${m.membership}</td>
            <td>${m.satisfaction}</td>
```

```
            <td>${m.dropoutFlag === 'Yes' ? '<span class="risk-badge">    High Risk</span>' :
'<span class="active-badge">    Active</span>'}</td>

            <td>${m.trainer}</td>

        </tr>

    `).join('');

  }



  // Show Member Profile

  window.showMemberProfile = function(id) {

    let m = fullMembers.find(x => x.id === id);

    alert(`    MEMBER PROFILE\n\n━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━\n    ID:
${m.id}\n    Name: ${m.name}\n    City: ${m.city}\n    Membership: ${m.membership}\n
Satisfaction: ${m.satisfaction}/5\n    Trainer: ${m.trainer}\n    Status: ${m.dropoutFlag ===
'Yes' ? '    Dropout Risk' : '    Active Member'}\n    Monthly Fee: ₹
${m.monthlyFee}\n━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━`);

  };



  // Apply Filters

  function applyFilters() {

    let search = document.getElementById('searchMember').value.toLowerCase();

    let city = document.getElementById('cityFilter').value;

    let membership = document.getElementById('membershipFilter').value;

    let risk = document.getElementById('riskFilter').value;



    currentFiltered = fullMembers.filter(m => {

      if (search && !m.id.toLowerCase().includes(search) &&
!m.name.toLowerCase().includes(search)) return false;

      if (city && m.city !== city) return false;
```

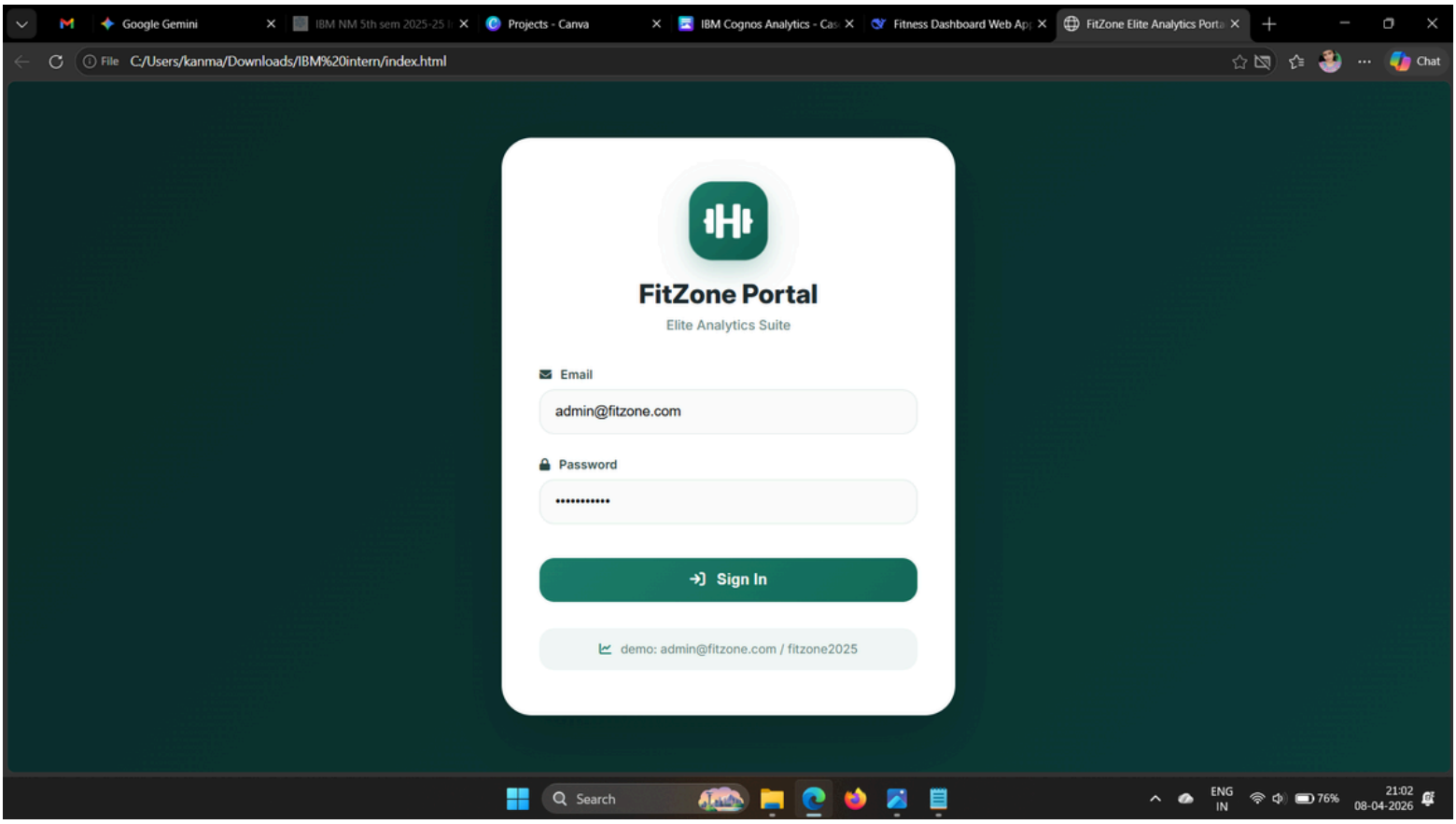
```
        if (membership && m.membership !== membership) return false;
        if (risk === 'high' && m.dropoutFlag !== 'Yes') return false;
        if (risk === 'active' && m.dropoutFlag !== 'No') return false;
        return true;
    });
    updateDashboard();
}

// Reset Filters
function resetFilters() {
    document.getElementById('searchMember').value = '';
    document.getElementById('cityFilter').value = '';
    document.getElementById('membershipFilter').value = '';
    document.getElementById('riskFilter').value = '';
    applyFilters();
}

// Export Functions
function exportToExcel() {
    let exportData = currentFiltered.map(m => ({
        'Member ID': m.id, 'Name': m.name, 'City': m.city, 'Membership': m.membership,
        'Satisfaction': m.satisfaction, 'Status': m.dropoutFlag, 'Trainer': m.trainer,
        'Monthly Fee': m.monthlyFee, 'Additional Revenue': Math.round(m.additionalRevenue)
    }));
    let ws = XLSX.utils.json_to_sheet(exportData);
    let wb = XLSX.utils.book_new();
```

```
    XLSX.utils.book_append_sheet(wb, ws, 'FitZone Members');
    XLSX.writeFile(wb, `FitZone_Report_${new Date().toISOString().slice(0,19)}.xlsx`);
}

function exportToPDF() {
    let element = document.querySelector('.main-content');
    html2pdf().from(element).set({ margin: 0.5, filename:
`FitZone_Dashboard_${Date.now()}.pdf` }).save();
}

// Dark Mode
function toggleDarkMode() {
    document.body.classList.toggle('dark-mode');
    localStorage.setItem('darkMode', document.body.classList.contains('dark-mode'));
}

// Check saved dark mode
if (localStorage.getItem('darkMode') === 'true') {
    document.body.classList.add('dark-mode');
}

// Event Listeners
document.getElementById('searchMember').addEventListener('input', applyFilters);
document.getElementById('cityFilter').addEventListener('change', applyFilters);
document.getElementById('membershipFilter').addEventListener('change', applyFilters);
document.getElementById('riskFilter').addEventListener('change', applyFilters);
```

```
document.getElementById('resetFilters').addEventListener('click', resetFilters);
document.getElementById('darkModeToggle').addEventListener('click', toggleDarkMode);
document.getElementById('exportExcel').addEventListener('click', exportToExcel);
document.getElementById('exportPdf').addEventListener('click', exportToPDF);

// Login Logic
document.getElementById('loginForm').addEventListener('submit', (e) => {
  e.preventDefault();
  let email = document.getElementById('loginEmail').value;
  let pwd = document.getElementById('loginPassword').value;
  if (email === 'admin@fitzone.com' && pwd === 'fitzone2025') {
    document.getElementById('loginWrapper').style.display = 'none';
    document.getElementById('dashboardContainer').style.display = 'block';
    applyFilters();
  } else {
    alert('Invalid credentials!\nUse: admin@fitzone.com / fitzone2025');
  }
});

document.getElementById('logoutBtn').addEventListener('click', () => {
  document.getElementById('dashboardContainer').style.display = 'none';
  document.getElementById('loginWrapper').style.display = 'flex';
});

// Initial load
applyFilters();
```

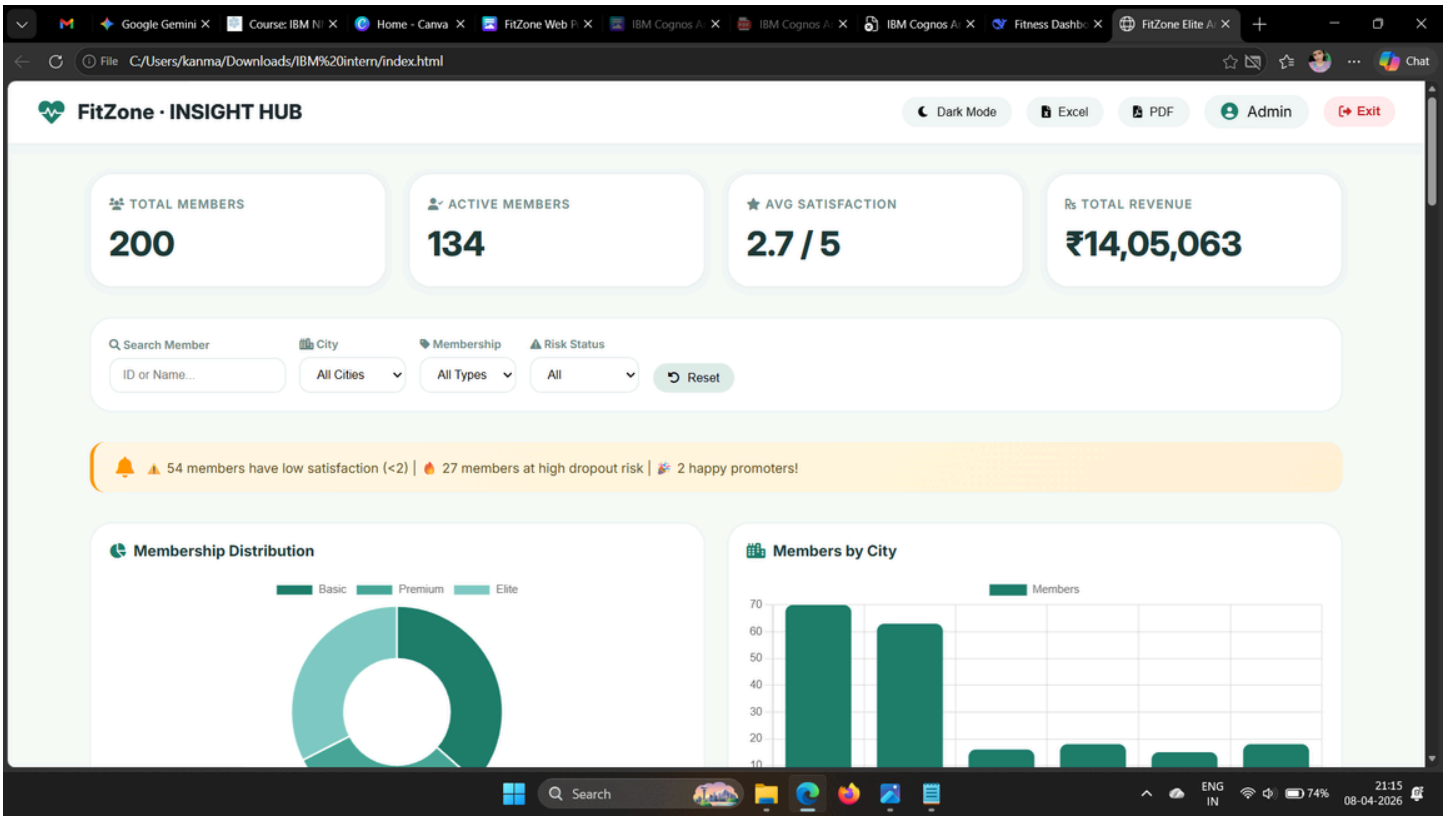
```
</script>

</body>

</html>
```

## FitZone Portal: Secure Administrator Login

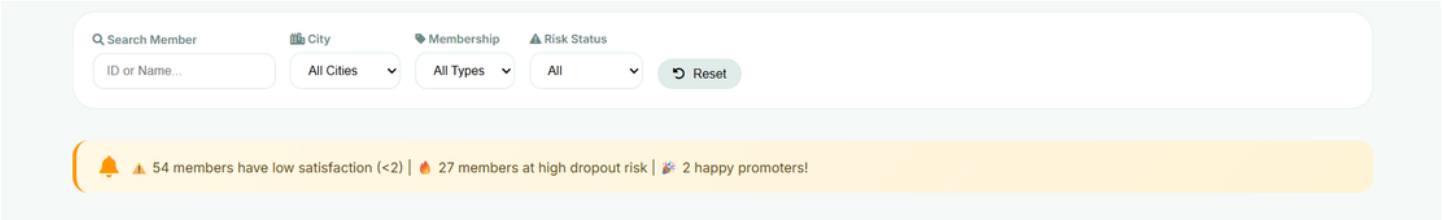



## Interactive Navigation & Action Controls

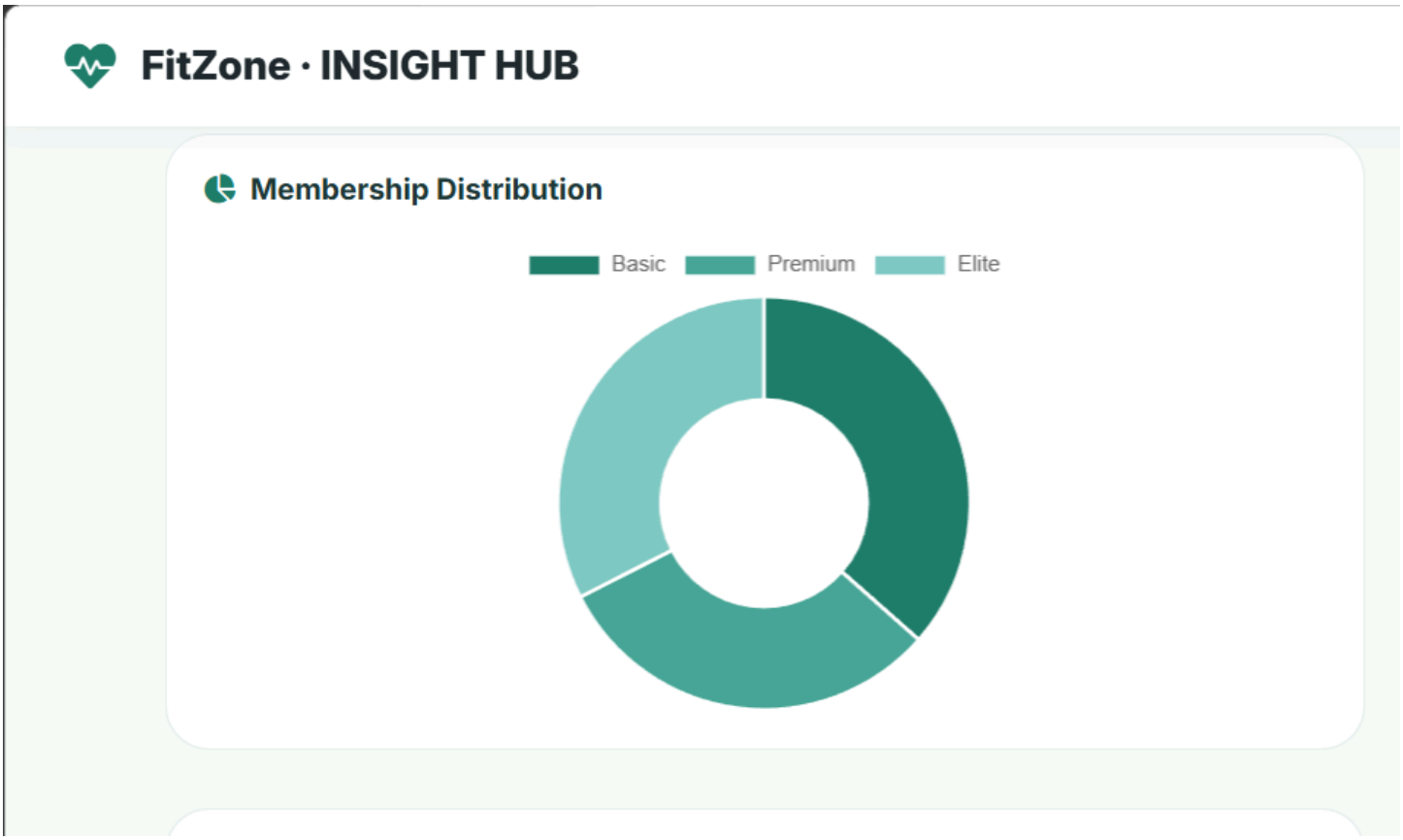
## FitZone Insight Hub: Live Analytics Dashboard



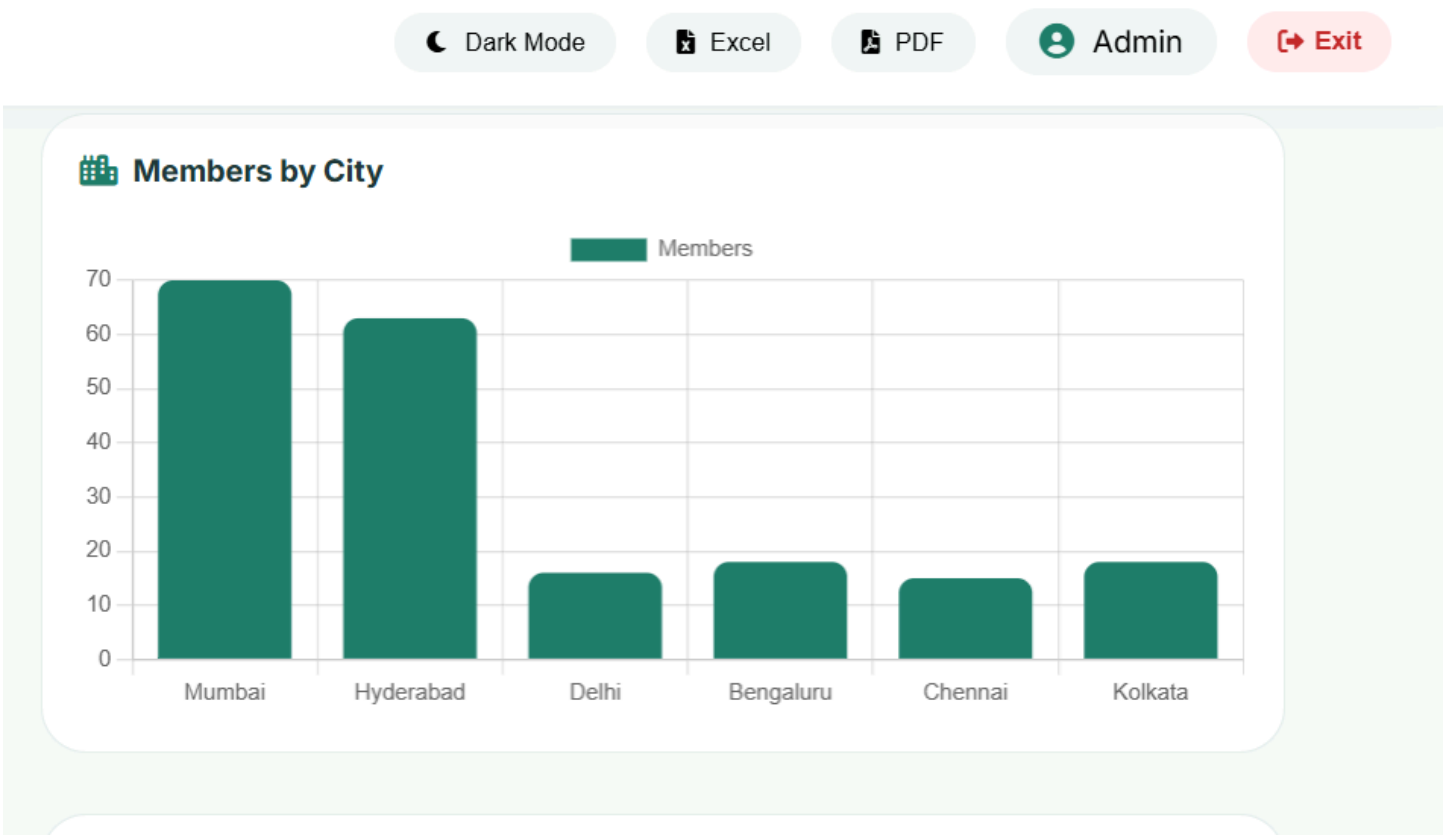

## Real-time Member Risk & Analytics Filters




## Visual Analysis: Membership Distribution

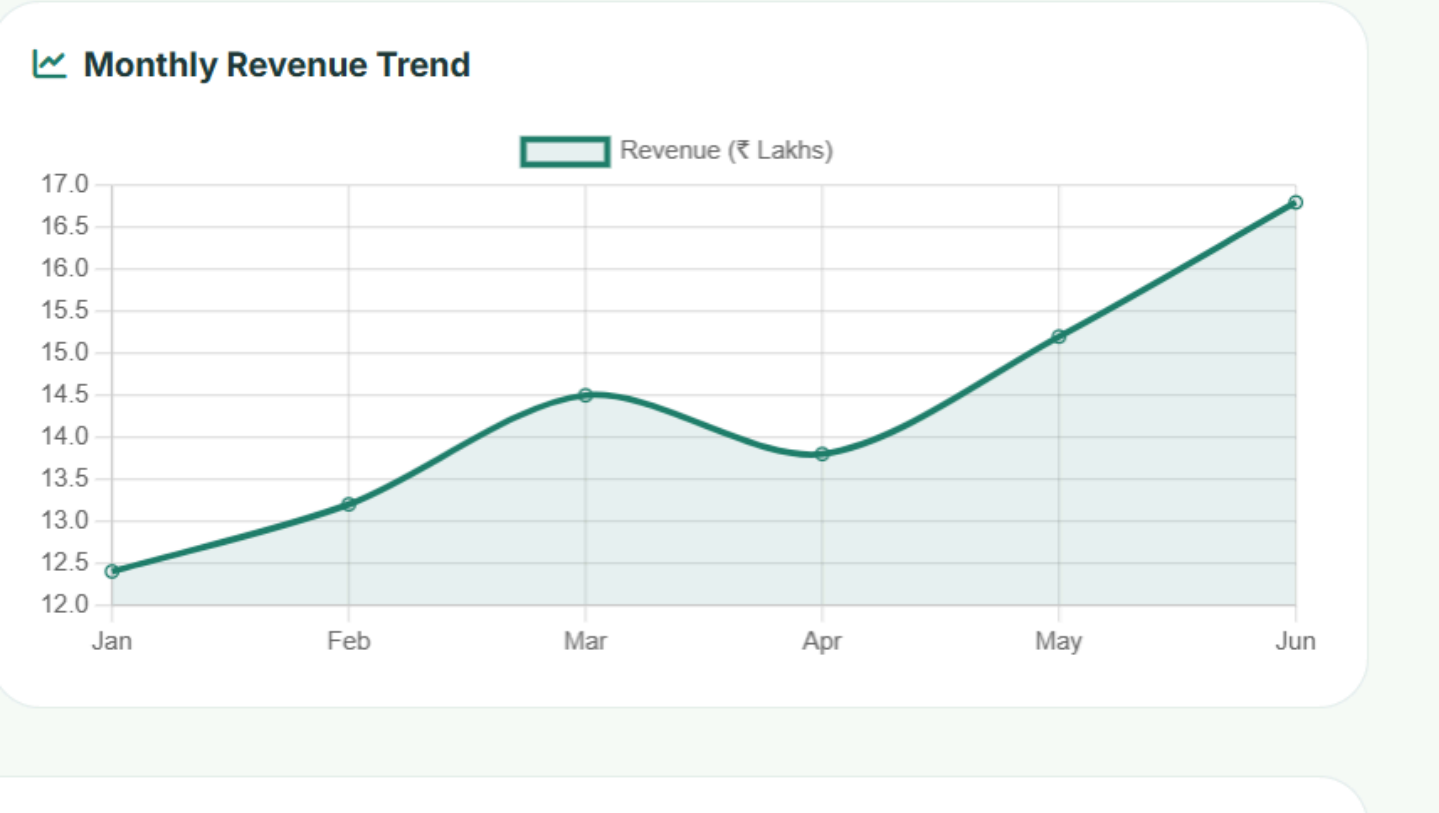



## Geographic Member Analysis




## Trainer Performance Metrics

## Monthly Revenue & Growth Analytics




## Member Directory (Dark Mode)

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Dark Mode | Excel | PDF | Admin | Exit

## Member Directory (click row for profile)

| Member ID | Name | City | Membership | Satisfaction | Status | Trainer |
|---|---|---|---|---|---|---|
| MEM0001 | Member_001 | Mumbai | Elite | 1.5 | Active | Trainer_Kiran |
| MEM0002 | Member_002 | Mumbai | Elite | 3.2 | Active | Trainer_Kiran |
| MEM0003 | Member_003 | Mumbai | Premium | 2.8 | Active | Trainer_Arjun |
| MEM0004 | Member_004 | Mumbai | Premium | 2.9 | Active | Trainer_Priya |
| MEM0005 | Member_005 | Mumbai | Basic | 1.4 | Active | Trainer_Ravi |
| MEM0006 | Member_006 | Mumbai | Elite | 1.1 | High Risk | Trainer_Ravi |
| MEM0007 | Member_007 | Mumbai | Basic | 2.7 | Active | Trainer_Kiran |
| MEM0008 | Member_008 | Mumbai | Premium | 3.5 | Active | Trainer_Priya |
| MEM0009 | Member_009 | Mumbai | Elite | 2.2 | Active | Trainer_Arjun |
| MEM0010 | Member_010 | Mumbai | Elite | 2.9 | High Risk | Trainer_Arjun |
| MEM0011 | Member_011 | Mumbai | Basic | 2.2 | Active | Trainer_Sunita |